An Empirical Study of Object Hallucination in BLIP 2: Multi Benchmark Evaluation and Visual Contrastive Decoding

Abstract

Vision Language Models (VLMs) frequently generate fluent descriptions not fully supported by the visual input, a phenomenon known as object hallucination. To empirically examine when and why visual grounding breaks down in compact architectures, we evaluate BLIP 2 OPT 2.7B across CHAIR, POPE, THRONE, and HallusionBench using 5,000 MS COCO images. We contrast free form captioning and binary object probing to isolate the effects of language priors, object co occurrence, parser sensitivity, and response bias. Furthermore, we implement Batched Visual Contrastive Decoding (VCD) as an inference time intervention designed to penalize over reliance on uni modal language priors. Through this unified evaluation framework, we investigate the trade offs between hallucination rates and object omissions, examine the alignment between discriminative probing and open ended generation, and analyze how contrastive decoding affects threshold calibration and visual grounding in smaller language backbones. Figure shows the input image.



1 Introduction

Large Vision Language Models (VLMs) often exhibit object hallucination: they generate fluent text asserting the presence of objects absent from the image. Prior work provides specialized benchmarks such as POPE for binary probing, CHAIR and THRONE for free form generation, and HallusionBench for diagnosing visual illusions versus language priors, alongside inference time mitigation methods such as Visual Contrastive Decoding (VCD). In this study, we systematically evaluate BLIP 2 on 5,000 MS COCO images across a comprehensive hallucination benchmark suite and examine the impact of VCD on confidence calibration, response bias, and the trade off between object hallucination and omission.

Which objects are hallucinated and in what context?
The "Furniture" super category accounts for 20% to 50% of all hallucinations, led by "dining table." Driven by linguistic and co occurrence priors, tokens like "sitting" or "top" frequently trigger "dining table." While hallucinations usually fit global scene context (e.g., a surfboard on a beach), they occasionally appear entirely out of context.

2 Methodology

2.1 CHAIR (Caption Hallucination Assessment with Image Relevance)

Object hallucination often arises from visual misclassification and an over reliance on language priors, causing models to memorize word co occurrences rather than grounding predictions in the actual image content. Because models can produce these fluent errors, standard sentence evaluation metrics often fail to adequately penalize hallucination. Furthermore, models that hallucinate heavily tend to make mistakes that are highly consistent with a standalone language model.

To quantify this phenomenon, the CHAIR metric calculates the proportion of generated words that represent actual objects in the image according to ground truth sentences and segmentations. Hallucination is

To quantify the model's probing performance, we calculate Precision, Accuracy, Recall, and the Yes Ratio:

$$\text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \quad \text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$

$$\text{Yes Ratio} = \frac{TP+FP}{TP+TN+FP+FN}, \quad \text{Brier Score} = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2 \tag{2}$$

measured using two variants:

CHAIR i (Instance level) evaluates the fraction of specific object instances that are hallucinated:

$$CHAIR_i = \frac{|\{\text{hallucinated objects}\}|}{|\{\text{all objectmentioned}\}|}$$

CHAIR s (Sentence level) evaluates the fraction of generated sentences that include at least one hallucinated object:

$$CHAIR_s = \frac{|\{\text{sentences with hallucinated object}\}|}{|\{\text{all sentences}\}|}$$

To implement this evaluation accurately, we follow the methodology of Rohrbach et al. by tokenizing each sentence and singularizing every word. A synonym mapping is then applied to align the generated terms with standard MSCOCO object categories. Crucially, the parser handles two word compounds (such as "hot dog") to ensure that standard 1 gram tokens (such as "dog") are not incorrectly flagged or extracted.

2.2 POPE (Polling based Object Probing Evaluation)

Disadvantages of CHAIR

Although CHAIR provides valuable insights into free form captioning, it is highly sensitive to prompt design; even semantically similar instructions can lead to substantially different metric values and alter the relative ranking of models. In addition, CHAIR relies on manually designed parsing rules to identify object mentions in generated captions. Furthermore, if a model like BLIP 2 generates a truncated output of only 8 words and stops, it is difficult to definitively state whether the model hallucinated or simply omitted objects from the scene.

Introduction to the POPE Score

To resolve the ambiguity of omissions versus hallucinations, we utilize a direct probing approach. By explicitly asking the model a binary question such as "Is there a [object] in the image?" we force a definitive "yes" or "no" prediction. This targeted probing yields a much more specific and accurate metric to evaluate object hallucination, which forms the basis for the POPE benchmark.

The Brier Score

Finally, we compute the Brier score to assess the confidence calibration of the model's responses. By extracting the first token probabilities for "yes" and "no", the Brier score measures the mean squared difference between the calibrated probability and the ground truth binary label, serving as a critical indicator of whether the model is inappropriately overconfident during a hallucination event.

2.3 THRONE (Text from image Hallucination Recognition with Object probes for open eNded Evaluation)

In LVLMs like BLIP 2, a Querying Transformer (Q Former) bridges a frozen image encoder with image and text pairs, while visual instruction tuning adapts the model for dialog. Although instruction tuning improves free form generation, it simultaneously worsens object hallucination.

THRONE evaluates these free form responses by distinguishing between **Type I hallucinations** (incorrect objects generated in open ended, free form descriptions) and **Type II hallucinations** (incorrect answers to binary yes or no object presence queries).

THRONE's Result on POPE. Evaluations show that Type I and Type II hallucinations are often disjoint, and LVLMs produce contradictory answers across both regimes. By employing public Language Models as abstractive question answering judges over generated text, THRONE measures Type I hallucinations while avoiding the restrictive parsing errors of CHAIR. Notably, gains on standard POPE benchmarks do not reliably reduce Type I hallucinations, and the two metrics are occasionally anti correlated.

Completing POPE for Type II Hallucinations (POPE C). Standard POPE systematically under samples negative object categories to artificially balance positive and negative queries, drastically underestimating true Type II hallucination rates. We therefore utilize POPE Complete (POPE C), which exhaustively queries the LVLM across every vocabulary class for all images. Evaluating on POPE C causes a severe F1 score drop compared to standard POPE, exposing the model's true vulnerability to Type II hallucinations.

2.4 HallusionBench

HallusionBench isolates visual reasoning failures into *Language Hallucination* (prioritizing parametric memory over visual input) and *Visual Illusion* (misinterpreting visual features), while quantifying affirmative "yes" bias via two metrics:

- **Yes Percentage Difference (d_y):** The deviation between predicted and ground truth "yes" rates. Values approaching 1 or minus 1 indicate extreme affirmative or negative bias, whereas an unbiased model scores near 0.0.
- **False Positive Ratio (r_{fp}):** The proportion of "yes" responses among all incorrect predictions. An unbiased model that avoids affirmative guessing scores near 0.5.

2.5 Visual Uncertainty and Contrastive Decoding

Object hallucinations emerge during auto regressive decoding $y_t \sim p_\theta(y_t \mid v, x, y_{<t})$ when uni modal LLM language priors and spurious MS COCO co occurrence biases overpower visual evidence. Our core intuition is that *degrading visual features forces the model to expose these internal biases*: under visual corruption, genuinely grounded object predictions lose confidence, whereas ungrounded tokens driven by text priors remain artificially probable. By applying T steps of forward Gaussian diffusion $q(v_t \mid v_{t-1}) = \mathcal{N}(v_t; \sqrt{1-\gamma}v_{t-1}, \gamma I)$ to the clean image v, we obtain a corrupted view $v' = v_T$ that strips spatial cues and isolates the model's ungrounded prior distribution.

Visual Contrastive Decoding (VCD) suppresses these amplified biases at inference time without parameter updates by subtracting logits of the noisy input v' from the clean input v:

$$p_{vcd}(y \mid v, v', x) = \text{softmax}\left((1+\alpha)\text{logit}_\theta(y \mid v, x) - \alpha \text{logit}_\theta(y \mid v', x)\right) \quad (3)$$

To avoid penalizing valid syntax invariant to visual input, an adaptive plausibility constraint restricts this update to a high confidence candidate set $\mathcal{V}_{head}(y_{<t}) = \{y_t \in \mathcal{V} : p_\theta(y_t \mid v, x, y_{<t}) \geq \beta \max_w p_\theta(w \mid v, x, y_{<t})\}$, zeroing out all other tokens. Empirical ablations confirm stable hallucination suppression and perception gains once $T > 500$ across $\alpha \in [0.25, 1.0]$, $\beta > 0$.

3 Results and Analysis

3.1 CHAIR

Table 1 shows that replacing naive matching with the enhanced parser reduces CHAIR_i from 2.72% to 2.06% and drops omissions by 14.79%, confirming that naive matching inflates false positives by penalizing valid synonyms and compound words. However, the high omission rate (59.16%) indicates that BLIP 2 minimizes hallucination primarily by adopting a conservative captioning strategy that sacrifices descriptive recall.

Table 1: CHAIR Benchmark Ablation on 5,000 Images (BLIP 2 OPT 2.7B)

Parser Variant	CHAIR i (%)	CHAIR s (%)	Omission (%)
Naive 1-gram (No Synonyms)	2.72	2.22	73.95
Enhanced (Synonyms+N-grams)	2.06	2.66	59.16

3.2 POPE

Table 2: POPE Quantitative & Calibration Results (BLIP 2 OPT 2.7B)

Split	Acc (%)	Prec (%)	Rec (%)	F1 (%)	Yes (%)	Brier
Random	75.79	74.16	79.16	76.58	53.37	0.2023
Popular	78.69	78.43	79.16	78.79	50.47	0.1927
Adversarial	74.25	72.08	79.16	75.46	54.91	0.2024

Table 2 shows performance degrades on the adversarial split (74.25% accuracy) due to an elevated "Yes" ratio (54.91%), indicating a vulnerability to co occurrence language priors. **Top 10 Hallucinated Objects**

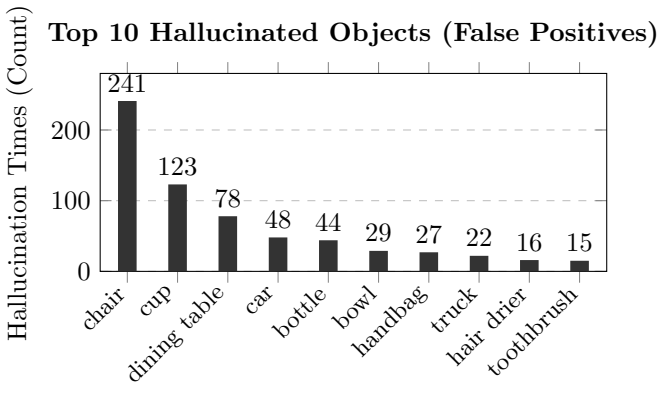


Figure 1: Distribution of top 10 hallucinated false positive classes on POPE. Frequent contextual objects like chairs and cups dominate the errors.

Qualitative analysis confirms the model frequently hallucinates common contextual items like chairs (241 times) and cups (123 times), while the overall Brier score (0.1991) reflects poor confidence calibration during these errors.

3.3 THRONE

Table 3: Combined Hallucination Evaluation (BLIP 2 OPT 2.7B)

Evaluation Split	Precision (%)	Recall (%)	F1 Score (%)
POPE (Random)	74.16	79.16	76.58
POPE (Popular)	78.43	79.16	78.79
POPE (Adversarial)	72.08	79.16	75.46
THRONE (Overall)	58.40	56.74	57.55
THRONE (Class wise)	67.15	58.81	62.70

Table 3 Synthesizes performance across both binary probing and open ended description generation. While BLIP 2 sustains moderate precision during targeted probing (POPE: 72%-78%), its performance degrades significantly during unconstrained generation (THRONE Class wise Precision: 67.15%, median response length 37 characters), demonstrating that free form contexts severely exacerbate object hallucination vulnerabilities. **The data proves that reducing or measuring Type II hallucinations (POPE) does not translate to a reduction in Type I hallucinations (THRONE). A model that appears highly accurate on POPE (76%+ F1) can still be highly prone to hallucinations in real world, open ended tasks (57% F1).**

3.4 HallusionBench

Table 4: HallusionBench Diagnostic Results (BLIP 2 OPT 2.7B)

aAcc (%)	qAcc (%)	fAcc (%)	d_y (Ideal $\sim$ 0)	r_{fp} (Ideal $\sim$ 0.5)
41.94	41.15	0.00	0.019	0.460

As presented in Table 4, the model exhibits balanced decision tendencies with a Yes Percentage Difference (d_y) of 0.019 and a False Positive Ratio (r_{fp}) of 0.460, showing near ideal resistance to affirmative bias. However, the 0.00% Figure Accuracy (fAcc) alongside a moderate All Accuracy (41.94%) reveals severe inconsistency under manipulated visual contexts, where the model consistently fails to answer all logically correlated questions for a given image correctly.

3.5 Visual Contrastive Decoding

Contrary to larger LVLMs, applying VCD to BLIP 2 OPT 2.7B slightly degrades POPE Random and Popular performance while remaining flat on the Adversarial split. Across splits, VCD elevates the affirmative response rate (reaching 55.32% on Random), lowering precision without yielding F1 gains. VCD assumes visual uncertainty amplifies strong uni modal LLM priors so they can be isolated and subtracted. However, the smaller OPT 2.7B backbone lacks overwhelming language bias; instead of penalizing prior driven hallucinations, contrastive subtraction destabilizes its fragile threshold calibration and amplifies baseline affirmative bias.

Table 5: POPE Benchmark Performance with Visual Contrastive Decoding (BLIP 2 OPT 2.7B)

Split	Acc (%)	Prec (%)	Rec (%)	F1 (%)	Yes (%)
Random	74.11	71.81	79.42	75.43	55.32
Popular	78.10	77.37	79.42	78.38	51.32
Adversarial	74.28	72.04	79.42	75.55	55.15

4 Conclusion

Despite its compact footprint and Q Former bottleneck, BLIP 2 resists the affirmative biases of larger instruction tuned models, scoring near ideal unbiased baselines on HallusionBench ($d_y = 0.019$, $r_{fp} = 0.46$) and accurately verifying constrained Type II POPE queries. However, BLIP 2 struggles with visual illusions and fine grained spatial grounding in open ended generation, where "Furniture" accounts for 20% to 50% of errors and contextual triggers (e.g., "sitting," "cat") induce co occurring hallucinations .

Addressing our core research questions, BLIP 2's measured hallucination rate depends heavily on task formulation, negative sampling density, and parser design. Binary probing (Type II) and free form generation (Type I) are largely disjoint and occasionally anti correlated: standard POPE inflates scores via negative undersampling whereas exhaustive POPE C exposes severe F1 drops, and THRONE's LM judging achieves 96% precision while cutting CHAIR's false discovery rate by over 50%. Finally, while degrading visual evidence ($T > 500$) forces larger LVLMs onto object frequency and co occurrence priors that VCD ($\beta > 0$, $\alpha \in [0.25, 1.0]$) and object enumeration effectively suppress, our OPT 2.7B findings prove contrastive decoding must be calibrated to backbone scale.

5 Key Takeaways

- Conservative captioning masks poor grounding behind high object omission rates.
- Binary object probing performance does not reflect open ended generation quality.
- Visual illusions and co occurrence priors drive most open ended hallucinations.

6 Refrence

- **Rohrbach et al. (2018)** — "Object Hallucination in Image Captioning," *EMNLP 2018* (CHAIR).
- **Li et al. (2023/2024)** — "Evaluating Object Hallucination in Large Vision Language Models," POPE; *ECCV 2024*.
- **Guan et al. (2024)** — "HallusionBench: An Advanced Diagnostic Suite for Entangled Language Hallucination and Visual Illusion in Large Vision Language Models," *CVPR 2024*.
- **Kaul et al. (2024)** — "THRONE: An Object based Hallucination Benchmark for the Free form Generations of Large Vision Language Models," *CVPR 2024*.
- **Leng et al. (2024)** — "Mitigating Object Hallucinations in Large Vision Language Models through Visual Contrastive Decoding," *CVPR 2024*.